

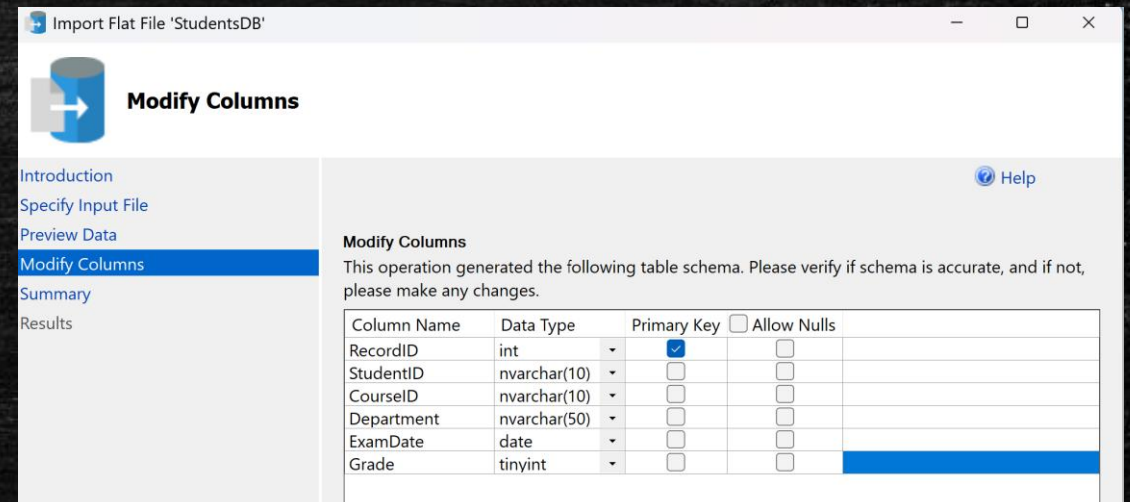
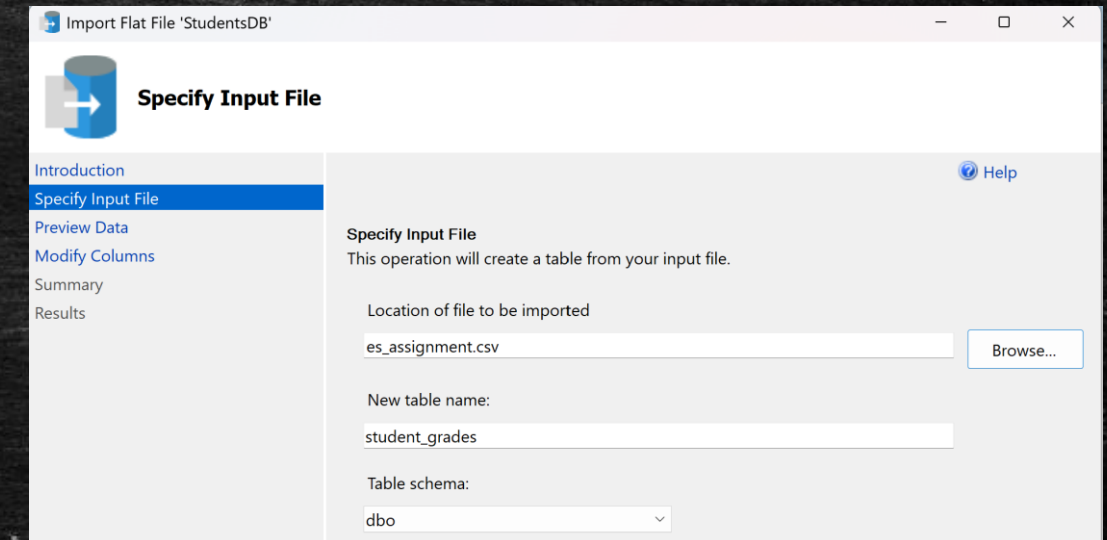
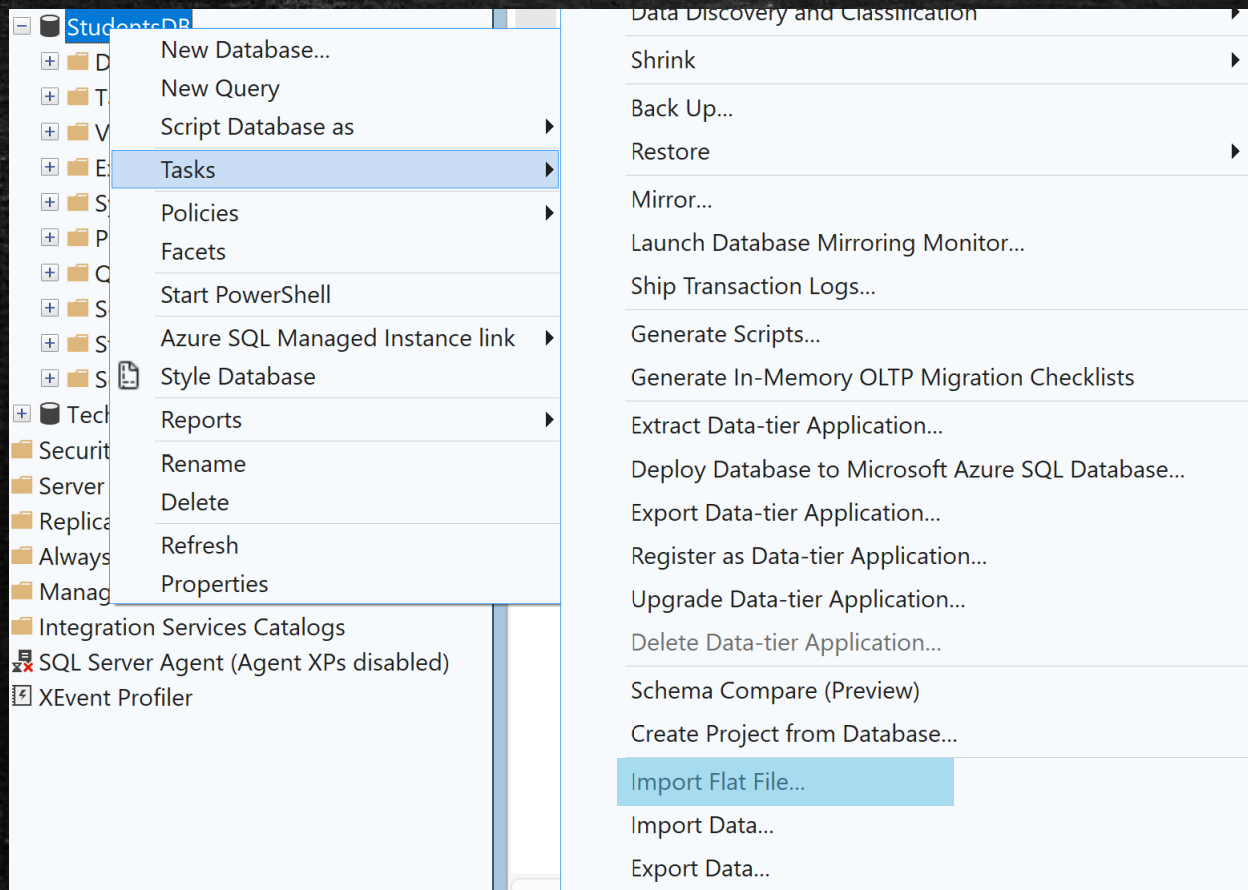
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Assignment 2 (SQL)

Load the provided dataset into your database

```
CREATE DATABASE StudentsDB;
```

```
USE StudentsDB;
```



Basic Questions

1. Query the total number of records.

```
SELECT COUNT(*) AS [Total Records]
FROM student_grades;
```

2. Query the average grade of all students.

```
SELECT AVG(Grade) AS [Average Grades]
FROM student_grades;
```

3. Query the highest grade obtained.

```
SELECT MAX(Grade) AS [Highest Grade]
FROM student_grades;
```

4. Query the lowest grade obtained.

```
SELECT MIN(Grade) AS [Lowest Grade]
FROM student_grades;
```

5. Query the number of distinct students.

```
SELECT COUNT(DISTINCT StudentID) AS [Total students]
FROM student_grades;
```


Basic Questions

6. Query the average grade for each course.

```
SELECT CourseID,  
        AVG(Grade) AS [Average Grades]  
FROM student_grades  
GROUP BY CourseID;
```

7. Query the number of students in each department.

```
SELECT Department,  
        COUNT(DISTINCT StudentID) AS [Total students]  
FROM student_grades  
GROUP BY Department;
```

8. Query the maximum grade for each course.

```
SELECT CourseID,  
        MAX(Grade) AS [Maximum Grade]  
FROM student_grades  
GROUP BY CourseID;
```

9. Query the minimum grade for each department.

```
SELECT Department,  
        MIN(Grade) AS [Minimum Grade]  
FROM student_grades  
GROUP BY Department;
```

10. Query the number of courses taken by each student.

```
SELECT StudentID,  
        COUNT(CourseID) AS [Total courses]  
FROM student_grades  
GROUP BY StudentID;
```


Intermediate Questions

11. Query courses where the average grade is greater than 75.

```
SELECT CourseID,  
        AVG(Grade) AS [Average Grades]  
FROM student_grades  
GROUP BY CourseID  
HAVING AVG(Grade) > 75;
```

12. Query students who have taken more than 3 courses.

```
SELECT StudentID,  
        COUNT(CourseID) AS [Total Courses]  
FROM student_grades  
GROUP BY StudentID  
HAVING COUNT(CourseID) > 3;
```

13. Query departments where the average grade is less than 65.

```
SELECT Department,  
        AVG(Grade) AS [Average Grades]  
FROM student_grades  
GROUP BY Department  
HAVING AVG(Grade) < 65;
```

14. Query students whose maximum grade is 100.

```
SELECT StudentID,  
        MAX(Grade) AS [Maximum Grade]  
FROM student_grades  
GROUP BY StudentID  
HAVING MAX(Grade) = 100;
```

15. Query courses that have been taken by at least 20 students.

```
SELECT CourseID,  
        COUNT(StudentID) AS [Total Students]  
FROM student_grades  
GROUP BY CourseID  
HAVING COUNT(*) >= 20;
```


Advanced Questions

16. Query students who have taken at least 4 courses and whose average grade is greater than 80.

```
SELECT StudentID,  
        COUNT(CourseID) AS [Total Courses],  
        AVG(Grade) AS [Average Grades]  
FROM student_grades  
GROUP BY StudentID  
HAVING COUNT(CourseID) >= 4  
        AND AVG(Grade) > 80;
```

17. Query departments where the maximum grade is at least twice the average grade.

```
SELECT Department,  
        MAX(Grade) AS [Maximum Grade],  
        AVG(Grade) AS [Average Grades]  
FROM student_grades  
GROUP BY Department  
HAVING MAX(Grade) >= 2 * AVG(Grade);
```

18. Query courses whose average grade is higher than the overall average grade.

```
SELECT CourseID,  
        AVG(Grade) AS [Average Grades]  
FROM student_grades  
GROUP BY CourseID  
HAVING AVG(Grade) > (SELECT AVG(Grade)  
                     FROM student_grades);
```

19 – 20. Query students whose lowest grade is greater than 70.

```
SELECT StudentID,  
        MIN(Grade) AS [Lowest Grade]  
FROM student_grades  
GROUP BY StudentID  
HAVING MIN(Grade) > 70;
```